

§1.28 — proto-morphisms and the idempotent-splitting category

Notes on Freyd & Scedrov, *Categories and Allegories* §1.28. Composition in **diagram order**: Ax means first A then x .

What a proto-morphism is (general, §1.2)

A category can be presented by a **two-sorted theory**: one sort for **objects**, one sort for **proto-morphisms**. A proto-morphism is a *raw arrow-datum* — a candidate arrow that does not yet have a fixed source/target.

What turns a proto-morphism into an actual morphism is the **source-target predicate** (a.k.a. arrow predicate) $A \xrightarrow{x} B$, read “ x may be construed as going from A to B ”. The *same* proto-morphism x may satisfy that predicate for several object pairs, or for none.

The category one actually obtains (§1.22) has as its **morphisms** the *instances* of the predicate — triples (A, x, B) , not the bare x .

proto-morphism = the underlying arrow data; morphism = that data together with a chosen source and target.

Convention: upper-case variables for objects, lower-case for proto-morphisms.

In §1.28 (the idempotent-splitting / Karoubi category)

§1.28 builds a category out of a class E of idempotents in a category A , presented via the §1.2–1.22 machinery:

- objects** = the idempotents in E ,
- proto-morphisms** = the morphisms of the original category A (reused as raw arrows),
- source-target predicate**: for idempotents A, B (upper-case = idempotents) and a proto-morphism x ,

$$A \xrightarrow{x} B \iff Ax = x = xB.$$

In diagram order: $A; x = x$ and $x; B = x$ — x is absorbed by A on the left and by B on the right.

So the proto-morphisms here are just **the arrows of A** , and the predicate $Ax = x = xB$ selects which of them count as morphisms between two idempotent-objects.

Consequences noted in the text

- The **identity** on an object A (an idempotent) is *A itself*: $A \cdot A = A = A \cdot A$ makes $A \xrightarrow{A} A$ hold, and A acts as the identity. This is why the forgetful map back to A is **not** generally a functor — when the idempotents are not all identities ($E \not\subseteq |A|$) it fails to preserve identities.
- Each idempotent $e \in E$ **splits** canonically in this category (§1.281): $e = xy$, $yx = 1$. That is the whole point — the Karoubi / idempotent-completion construction. (§1.282: any two splittings of the same idempotent are canonically isomorphic.)

Which side is the fixed point?

It is x that is the fixed point; A and B are the idempotents acting *on* it. Read $Ax = x = xB$ as two fixed-point conditions on x :

- $Ax = x$ — x is fixed by A acting **on the left** (pre-composition). A behaves like an identity for x on the **source** side.
- $xB = x$ — x is fixed by B acting **on the right** (post-composition). B behaves like an identity for x on the **target** side.

So x is a fixed point of *both* operators “left-multiply by A ” and “right-multiply by B ”. The idempotents A, B are the would-be identities; x is the arrow they fix from either side — which is exactly what makes A the left identity (source) and B the right identity (target) of x in the new category. (Saying “ A is x ’s fixed point” reverses the roles: A is the operator, x is what is held fixed.)

Why this is element-free, and why that matters

The presentation is **arrows-only**: A and x live in the *same* sort (proto-morphisms), so both Ax and xB are just composition — you can multiply on the **left** and on the **right**. If x were an *element* of a set and A a function, you would only have one-sided application $x(A)$, and there would be no way to say “fix x from the right.” The two-sided absorption is only expressible because everything is an arrow.

This is exactly why categories generalize monoids, and why objects can be **identified with their identity arrows**: an identity is characterized purely equationally by $1x = x = x1$, with no reference to what x “does” to anything. Source and target are recovered as *the* left / right absorbing idempotents — no elements, no application.

§1.28 is the payoff of taking that seriously: since objects are just two-sided identities, any idempotent that *acts* like a two-sided identity ($Ax = x = xB$) is allowed to *be* an object. Drop the requirement that identities be “the real” ones and let every idempotent in E serve as an object. The element-free formulation is what makes that move even expressible.

Does every idempotent split? (and is splitting an AC property?)

Idempotent splitting is **not** automatic, and it has **nothing to do with AC regularity**.

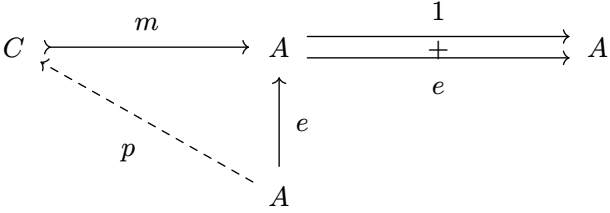
An idempotent $e : A \rightarrow A$ *splits* if there are $x : A \rightarrow B$, $y : B \rightarrow A$ with $xy = e$ and $yx = 1_B$. A category in which all idempotents split is called **idempotent-complete** (a.k.a. Cauchy / Karoubi complete) — a special property, not something a general category has.

The $\text{Split}(E)$ construction of §1.28 is exactly the device that *forces* splitting: adjoin the idempotents as objects, and each $e \in E$ then splits canonically (§1.281). If $E =$ all idempotents, all idempotents split in the completed category. (§1.573 does precisely this for the primitive-recursive functions: that category lacks equalizers, so idempotents are split by hand to build P .)

But in a Cartesian category they split for free — no AC needed

Any category with equalizers splits all of its idempotents, and a Cartesian category has equalizers. The construction is the one used in `ac_factorization_via_idempotent` (§1.571):

Given an idempotent $e : A \rightarrow A$, let $m : C \rightarrow A$ be the equalizer of 1_A and e . Since $e; e = e = e; 1$, the map e *equalizes* the parallel pair $1, e$, hence factors **uniquely** through the equalizer: there is a unique $p : A \rightarrow C$ with $p; m = e$ (unique because m is monic).



The apex A is a cone over $1, e$ via the vertical leg e (since $e; 1 = e = e; e$); it factors uniquely through the equalizer m by the dashed p , with $p; m = e$.

Then

- $m; p = 1_C$ (because $m; p; m = m; e = m = 1$; m and m is monic), and
- $p; m = e$.

So e splits as (p, m) , with p left-invertible. **This uses only the equalizer, i.e. Cartesian-ness — no AC, no “same level”**. And an equalizer *is a limit* (of the parallel-pair diagram $A \rightrightarrows A$), so this comes packaged with the finite-limit structure of a Cartesian category.

Where AC actually comes in

Splitting alone gives the *left-invertible* leg p . Being **AC regular** (§1.57) is stronger: *every* morphism must factor as left-invertible ; **monic**. The §1.571 hypothesis (an idempotent e with $ex = x$ and the same level as x) is what upgrades “idempotents split” to “AC regular”: the splitting handles p , and the same-level condition is the extra ingredient forcing the second leg $n = m; x$ to be **monic** (see `Fredy/S1_57.md`).

Property	What it needs
Every idempotent splits	equalizers ($\Leftarrow$ Cartesian), or do §1.28’s $\text{Split}(E)$ by hand
AC regular	the above plus §1.571’s same-level idempotents (the AC content)